

DESIGN AND IMPLEMENTATION OF SPEED CONTROL OF BLDC MOTOR

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DOI : <https://www.doi.org/10.56726/IRJMETS24969>

ABSTRACT

Brushless DC motors are displacing brushed motors in a variety of industrial applications. Electric vehicles with BLDC motors are gaining popularity in transportation as well. Electronic commutation gives BLDC motors several benefits over brushed motors with mechanical commutation. Because of the electronic commutation, a controller is required to turn on and operate the motor. The torque can also be controlled with the controller. The rotor of a BLDC motor is a permanent magnet, and the stator is made up of coiled windings. The stator's back EMF is trapezoidal. The position of the rotor is determined by sensing the back EMF using an EMF observer. For speed control, the proposed methodology uses a PIC microcontroller. The BLDC motor is driven by power MOSFETs. MATLAB Simulink is used to simulate the same with advanced features like hysteresis current controller. Regenerative braking, in which the excess energy created is fed back to the storage device, will be implemented in the future.

Keywords: Brushless, BLDC, Speed Control, Electronic Commutation, Back EMF, Regeneration.

I. INTRODUCTION

Electric vehicles are becoming more popular in India. Electric vehicles are manufactured by a few companies, while components are supplied by others. Imported vehicle controllers make up the great majority of the market. This is because electric vehicles and BLDC motors are relatively new in India, where induction motors and brushed DC motors are commonly used and do not require controllers.

II. METHODOLOGY

The proposed method comprises regulating the speed of a sensorless Brushless DC motor whose rotor position is governed by the back EMF it produces. This is a working prototype of a controller that can control electric vehicles. A back EMF observer detects the back EMF, which transmits signals to the PIC microcontroller. These signals contain information of the rotor's position. PWM signals are generated by the PIC microcontroller and supplied to the MOSFET driver circuit. Signals are created, and particular phases are turned on based on the rotor's position. The motor speed can be regulated by altering the width of the PWM pulses.

III. MODELING AND ANALYSIS

The proposed system's block diagram is as follows. MOSFETs, whose gates are actuated according to the rotor position, make up the electronic commutation. The Back EMF observer provides this information. For each phase of the motor, there are a pair of MOSFETs. By varying the rheostat attached to the PIC microcontroller, the PWM width and hence the speed of the motor can be controlled.

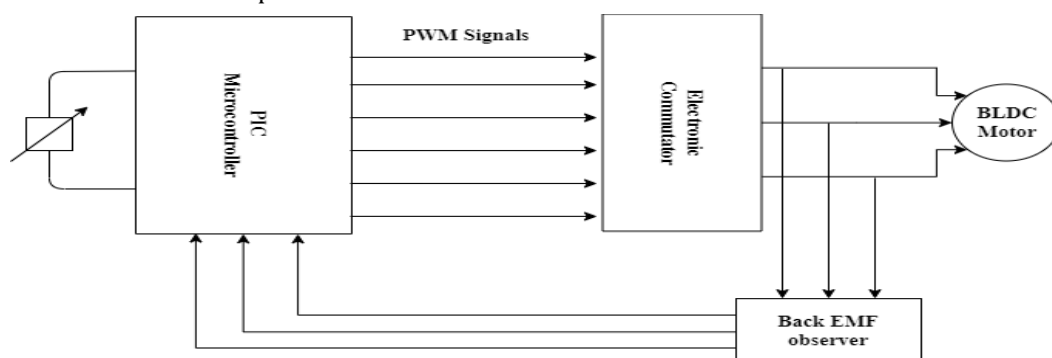


Figure 1: Block diagram of proposed system

The process can be indicated by the following flowchart.

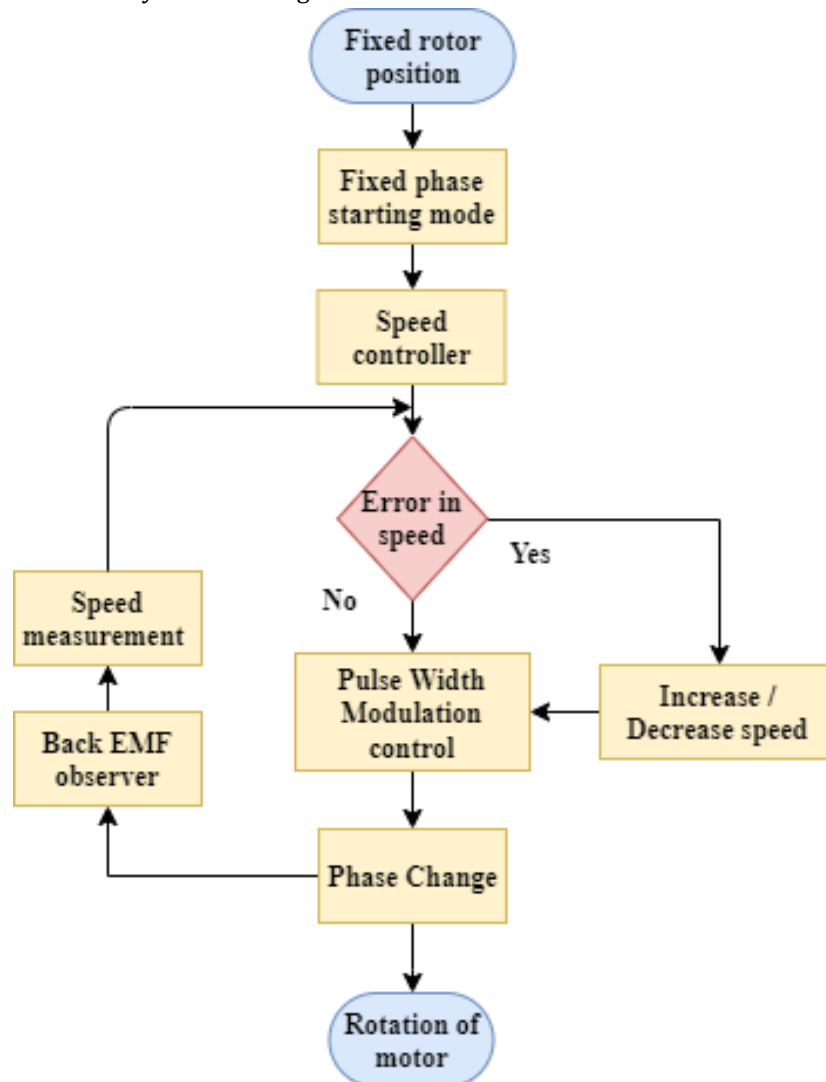


Figure 2: Flow chart of the process

The simulation has been carried out in MATLAB Simulink using complex control systems that are planned to be implemented in real-world system. The Simulation has been done by creating subsystems for Back EMF observer, speed calculator, commutation signals, current wave generator, speed converter, hysteresis current controller, inverter and parameters to be measured. The stator current waveforms, EMF and the speed of the motor are observed using different scopes. The entire speed control mechanism as done in Simulink is shown in the diagram below.

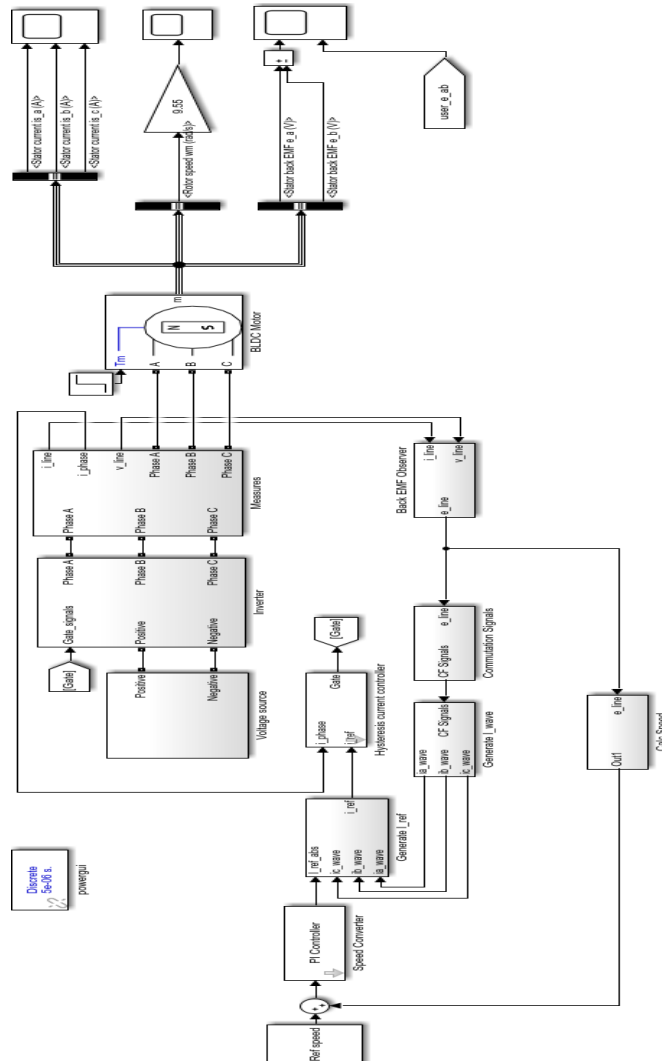


Figure 3: Simulink diagram of speed control system

The stator current waveforms as measured are shown in the first scope's output below.

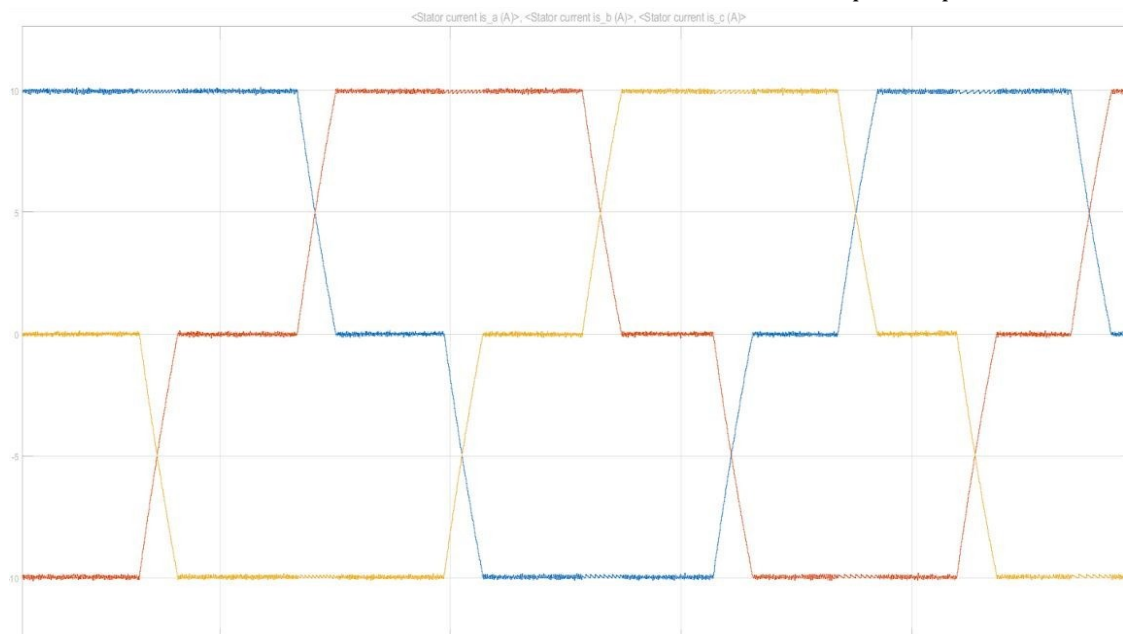


Figure 4: Stator current waveforms

The rotor speed was set to 3500 rpm, and the same result was obtained, as seen below.

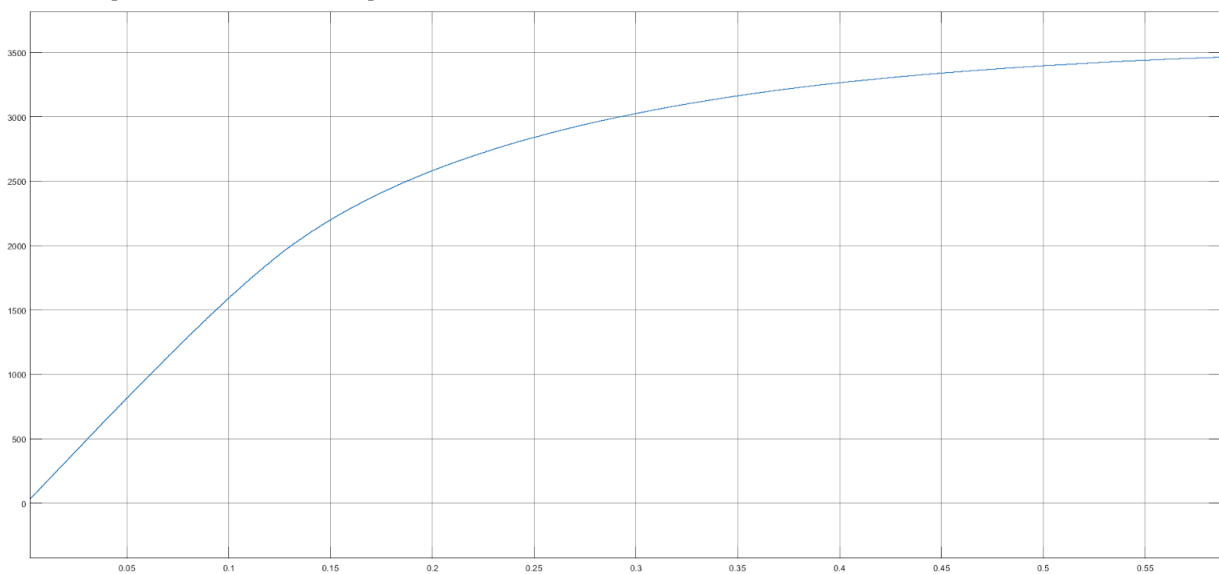


Figure 5: Motor speed

The ideal and real EMFs are shown in the following output of the third scope. The ideal EMF is trapezoidal; however the actual EMF is nearly trapezoidal but not entirely trapezoidal.

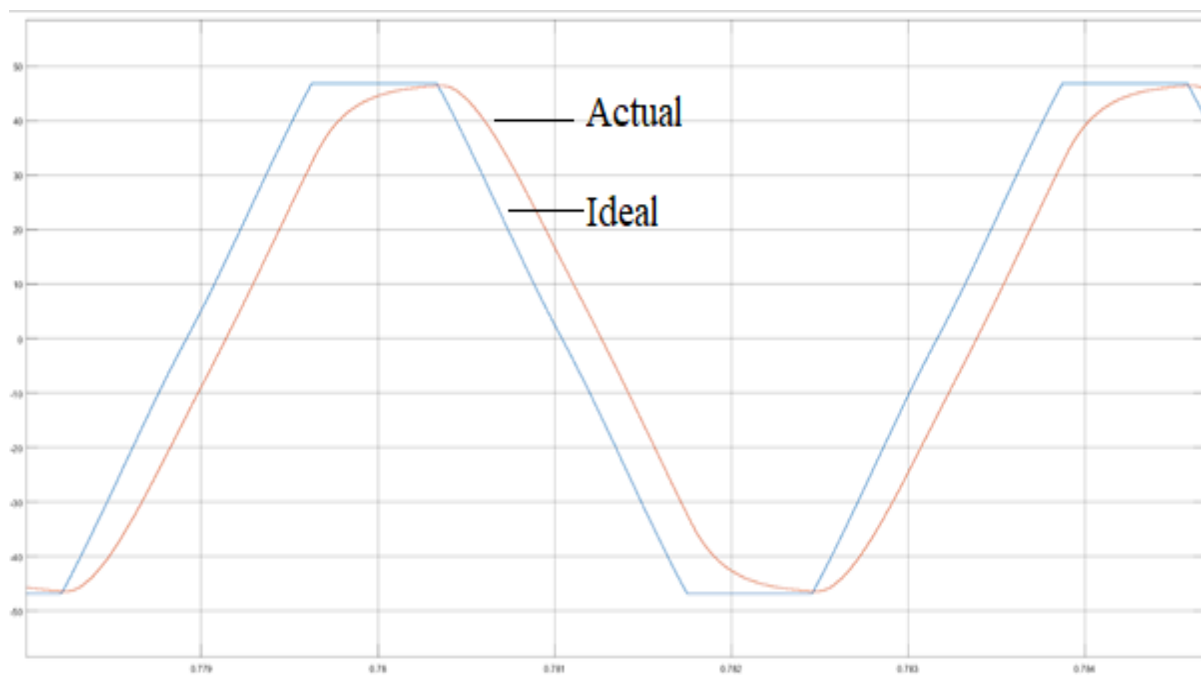


Figure 6: Ideal and actual EMF waveforms

IV. RESULT

EMF observer was utilized to control the speed of the BLDC motor. Advanced speed control has been proposed, and it has been simulated with MATLAB Simulink. In addition, regenerative braking is being considered for this project. This aids in the development of a controller with sophisticated and improved control as well as regenerative braking, which uses bidirectional converters to convert frictional energy spent while applying brakes to electrical energy.

V. CONCLUSION

As a result, a speed controller prototype for BLDC motors has been created. This prototype contains some basic speed control functions, and the tests were successful. Developing and manufacturing converters in India will help to provide new job opportunities and a new growth sector for trained workers.

VI. REFERENCES

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